

AIR POLLUTION AND CPCB STANDARDS

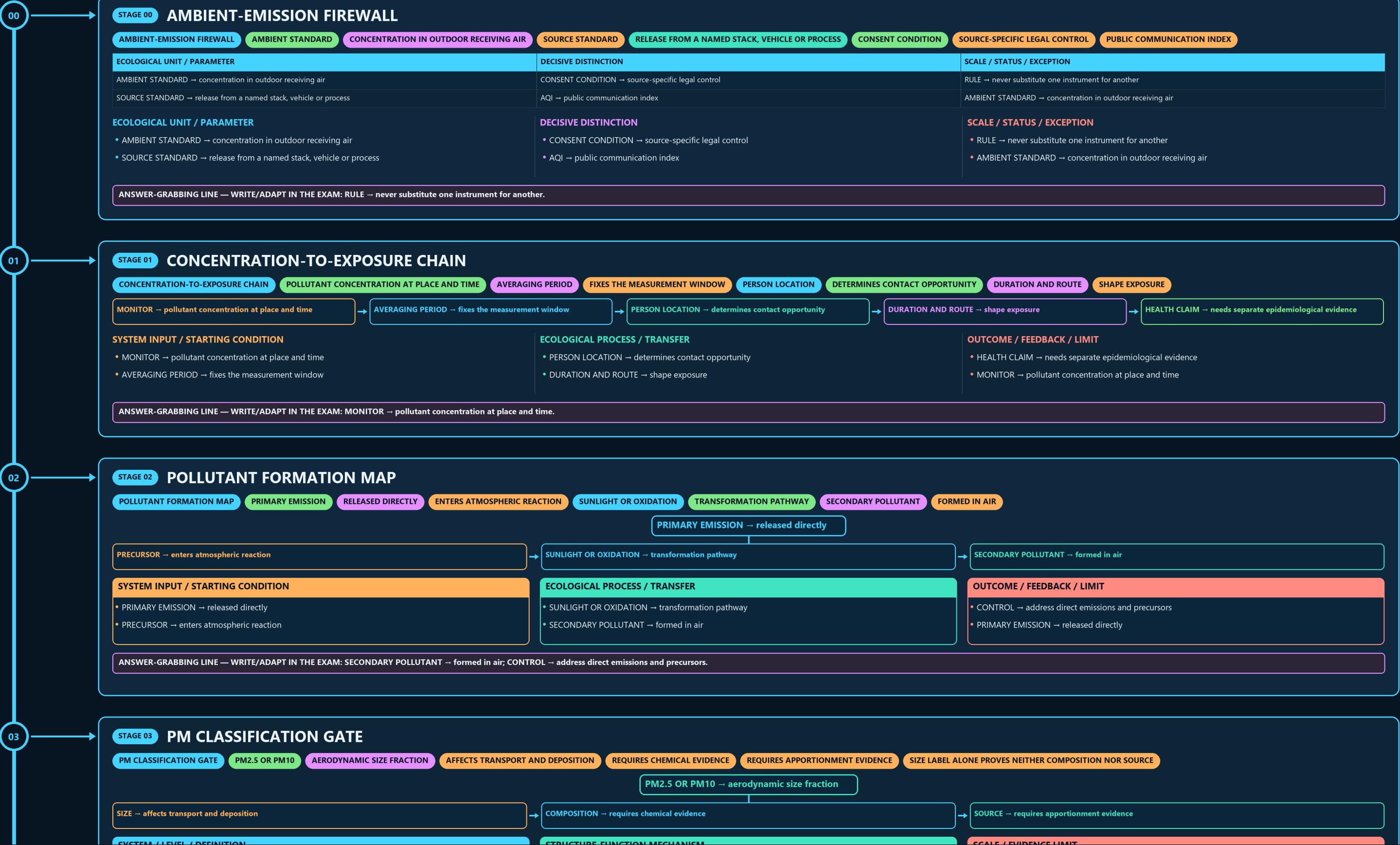
AMBIENT-EMISSION FIREWALL → CONCENTRATION-TO-EXPOSURE CHAIN → POLLUTANT FORMATION MAP → PM CLASSIFICATION GATE → SMOG AND ACID-DEPOSITION MATRIX → NAAQS DATA GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

PRECURSOR → enters atmospheric reaction

SECONDARY POLLUTANT → formed in air

PRIMARY POLLUTANT → released directly

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECONDARY POLLUTANT → formed in air; CONTROL → address direct emissions and precursors.

STAGE 03

PM CLASSIFICATION GATE

PM CLASSIFICATION GATEPM2.5 OR PM10AERODYNAMIC SIZE FRACTIONAFFECTS TRANSPORT AND DEPOSITIONREQUIRES CHEMICAL EVIDENCEREQUIRES APPORTIONMENT EVIDENCESIZE LABEL ALONE PROVES NEITHER COMPOSITION NOR SOURCE

PM2.5 OR PM10 → aerodynamic size fraction

SIZE → affects transport and deposition

COMPOSITION → requires chemical evidence

SOURCE → requires apportionment evidence

SYSTEM / LEVEL / DEFINITION

PM2.5 OR PM10 → aerodynamic size fraction

SIZE → affects transport and deposition

STRUCTURE-FUNCTION MECHANISM

COMPOSITION → requires chemical evidence

SOURCE → requires apportionment evidence

SCALE / EVIDENCE LIMIT

TRAP → size label alone proves neither composition nor source

PM2.5 OR PM10 → aerodynamic size fraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → size label alone proves neither composition nor source.

04

STAGE 04

SMOG AND ACID-DEPOSITION MATRIX

SMOG AND ACID-DEPOSITION MATRIXPRECURSOR EMISSIONS PLUS SUNLIGHT-DRIVEN CHEMISTRYSURFACE OZONESECONDARY OXIDANTACID DEPOSITIONSULPHUR AND NITROGEN PRECURSOR PATHWAYWET OR DRYDEPOSITION ROUTES

| ECOLOGICAL UNIT / PARAMETER                               | DECISIVE DISTINCTION                                     | SCALE / STATUS / EXCEPTION                                |
|-----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------|
| SMOG → precursor emissions plus sunlight-driven chemistry | ACID DEPOSITION → sulphur and nitrogen precursor pathway | RULE → precursor is not the final atmospheric product     |
| SURFACE OZONE → secondary oxidant                         | WET OR DRY → deposition routes                           | SMOG → precursor emissions plus sunlight-driven chemistry |

ECOLOGICAL UNIT / PARAMETER

SMOG → precursor emissions plus sunlight-driven chemistry

SURFACE OZONE → secondary oxidant

DECISIVE DISTINCTION

ACID DEPOSITION → sulphur and nitrogen precursor pathway

WET OR DRY → deposition routes

SCALE / STATUS / EXCEPTION

RULE → precursor is not the final atmospheric product

SMOG → precursor emissions plus sunlight-driven chemistry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → precursor is not the final atmospheric product.

05

STAGE 05

NAAQS DATA GATE

NAAQS DATA GATEIDENTIFY EXACT SUBSTANCE OR FRACTIONPRESERVE THE NOTIFIED UNITAVERAGING PERIODANNUAL, DAILY OR SHORTER ONLY IF SOURCEDAREA AND COMPLIANCE RULEREAD NOTIFICATIONNO TABLE

POLLUTANT → identify exact substance or fraction

UNIT → preserve the notified unit

AVERAGING PERIOD → annual, daily or shorter only if sourced

AREA AND COMPLIANCE RULE → read notification

SYSTEM / LEVEL / DEFINITION

POLLUTANT → identify exact substance or fraction

UNIT → preserve the notified unit

STRUCTURE-FUNCTION MECHANISM

AVERAGING PERIOD → annual, daily or shorter only if sourced

AREA AND COMPLIANCE RULE → read notification

SCALE / EVIDENCE LIMIT

NO TABLE → no invented limit

POLLUTANT → identify exact substance or fraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AVERAGING PERIOD → annual, daily or shorter only if sourced.

06

STAGE 06

AQI STATUTORY BOUNDARY

AQI STATUTORY BOUNDARYMONITORED POLLUTANT CONCENTRATIONSPUBLIC-FACING CATEGORYNOTIFIED AMBIENT STANDARDSOURCE LIMITSEPARATE EMISSION CONTROLAQI CATEGORY IS NOT EITHER STANDARD TABLE

| ECOLOGICAL UNIT / PARAMETER                | DECISIVE DISTINCTION                     | SCALE / STATUS / EXCEPTION                          |
|--------------------------------------------|------------------------------------------|-----------------------------------------------------|
| INPUT → monitored pollutant concentrations | NAAQS → notified ambient standard        | VERDICT → AQI category is not either standard table |
| OUTPUT → public-facing category            | SOURCE LIMIT → separate emission control | INPUT → monitored pollutant concentrations          |

ECOLOGICAL UNIT / PARAMETER

INPUT → monitored pollutant concentrations

DECISIVE DISTINCTION

NAAQS → notified ambient standard

SCALE / STATUS / EXCEPTION

VERDICT → AQI category is not either standard table

AIR POLLUTION AND CPCB STANDARDS • TILE 2/4 • same master rows 2533-5767 of 10834 • continues below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AVERAGING PERIOD → annual, daily or shorter only if sourced.

06

STAGE 06 AQI STATUTORY BOUNDARY

AQI STATUTORY BOUNDARY MONITORED POLLUTANT CONCENTRATIONS PUBLIC-FACING CATEGORY NOTIFIED AMBIENT STANDARD SOURCE LIMIT SEPARATE EMISSION CONTROL AQI CATEGORY IS NOT EITHER STANDARD TABLE

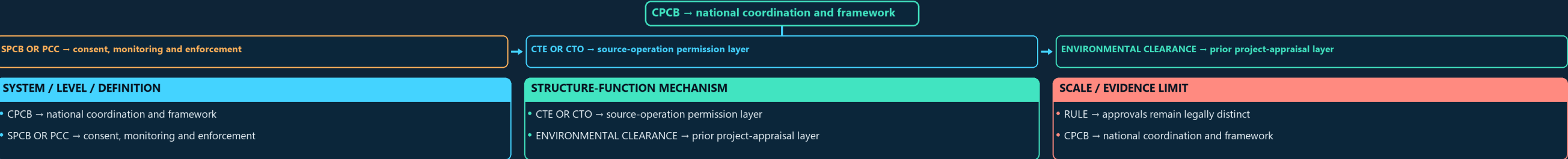
| ECOLOGICAL UNIT / PARAMETER                                                                                                                                           | DECISIVE DISTINCTION                                                                                                                                           | SCALE / STATUS / EXCEPTION                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT → monitored pollutant concentrations                                                                                                                            | NAAQS → notified ambient standard                                                                                                                              | VERDICT → AQI category is not either standard table                                                                                                                                      |
| OUTPUT → public-facing category                                                                                                                                       | SOURCE LIMIT → separate emission control                                                                                                                       | INPUT → monitored pollutant concentrations                                                                                                                                               |
| <b>ECOLOGICAL UNIT / PARAMETER</b> <ul style="list-style-type: none"><li>INPUT → monitored pollutant concentrations</li><li>OUTPUT → public-facing category</li></ul> | <b>DECISIVE DISTINCTION</b> <ul style="list-style-type: none"><li>NAAQS → notified ambient standard</li><li>SOURCE LIMIT → separate emission control</li></ul> | <b>SCALE / STATUS / EXCEPTION</b> <ul style="list-style-type: none"><li>VERDICT → AQI category is not either standard table</li><li>INPUT → monitored pollutant concentrations</li></ul> |

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → AQI category is not either standard table.

07

STAGE 07 INSTITUTION AND CLEARANCE LADDER

INSTITUTION AND CLEARANCE LADDER NATIONAL COORDINATION AND FRAMEWORK SPCB OR PCC CONSENT, MONITORING AND ENFORCEMENT CTE OR CTO SOURCE-OPERATION PERMISSION LAYER ENVIRONMENTAL CLEARANCE PRIOR PROJECT-APPRAISAL LAYER

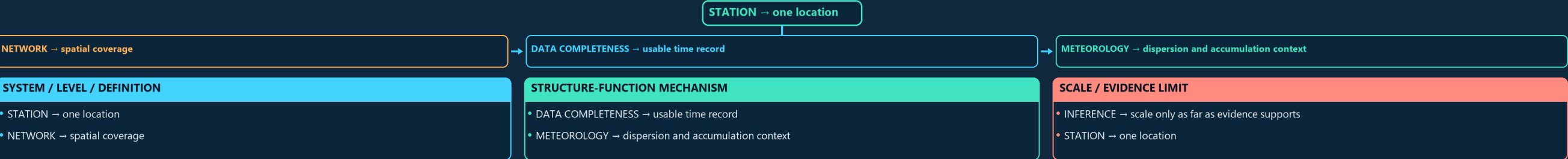


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPCB OR PCC → consent, monitoring and enforcement.

08

STAGE 08 MONITORING EVIDENCE LADDER

MONITORING EVIDENCE LADDER ONE LOCATION SPATIAL COVERAGE DATA COMPLETENESS USABLE TIME RECORD DISPERSION AND ACCUMULATION CONTEXT SCALE ONLY AS FAR AS EVIDENCE SUPPORTS



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INFERENCE → scale only as far as evidence supports.

09

STAGE 09 SOURCE-APPORTIONMENT CUBE

SOURCE-APPORTIONMENT CUBE SOURCE MIX CHANGES BY LOCATION ACTIVITY AND METEOROLOGY CHANGE PM, OZONE AND GASES HAVE DIFFERENT PATHWAYS MODEL AND SAMPLING ASSUMPTIONS MATTER NO UNIVERSAL SHARE DATE AND SCOPE EVERY ESTIMATE

| ECOLOGICAL UNIT / PARAMETER                                                                                                                                               | DECISIVE DISTINCTION                                                                                                                                                                       | SCALE / STATUS / EXCEPTION                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CITY → source mix changes by location                                                                                                                                     | POLLUTANT → PM, ozone and gases have different pathways                                                                                                                                    | NO UNIVERSAL SHARE → date and scope every estimate                                                                                                                                 |
| SEASON → activity and meteorology change                                                                                                                                  | METHOD → model and sampling assumptions matter                                                                                                                                             | CITY → source mix changes by location                                                                                                                                              |
| <b>ECOLOGICAL UNIT / PARAMETER</b> <ul style="list-style-type: none"><li>CITY → source mix changes by location</li><li>SEASON → activity and meteorology change</li></ul> | <b>DECISIVE DISTINCTION</b> <ul style="list-style-type: none"><li>POLLUTANT → PM, ozone and gases have different pathways</li><li>METHOD → model and sampling assumptions matter</li></ul> | <b>SCALE / STATUS / EXCEPTION</b> <ul style="list-style-type: none"><li>NO UNIVERSAL SHARE → date and scope every estimate</li><li>CITY → source mix changes by location</li></ul> |

09

STAGE 09 SOURCE-APPORTIONMENT CUBE

SOURCE-APPORTIONMENT CUBE SOURCE MIX CHANGES BY LOCATION ACTIVITY AND METEOROLOGY CHANGE PM, OZONE AND GASES HAVE DIFFERENT PATHWAYS MODEL AND SAMPLING ASSUMPTIONS MATTER NO UNIVERSAL SHARE DATE AND SCOPE EVERY ESTIMATE

| ECOLOGICAL UNIT / PARAMETER              | DECISIVE DISTINCTION                                    | SCALE / STATUS / EXCEPTION                         |
|------------------------------------------|---------------------------------------------------------|----------------------------------------------------|
| CITY → source mix changes by location    | POLLUTANT → PM, ozone and gases have different pathways | NO UNIVERSAL SHARE → date and scope every estimate |
| SEASON → activity and meteorology change | METHOD → model and sampling assumptions matter          | CITY → source mix changes by location              |

ECOLOGICAL UNIT / PARAMETER

- CITY → source mix changes by location
- SEASON → activity and meteorology change

DECISIVE DISTINCTION

- POLLUTANT → PM, ozone and gases have different pathways
- METHOD → model and sampling assumptions matter

SCALE / STATUS / EXCEPTION

- NO UNIVERSAL SHARE → date and scope every estimate
- CITY → source mix changes by location

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO UNIVERSAL SHARE → date and scope every estimate.

10

STAGE 10 PROGRAMME-RESPONSE ARCHITECTURE

PROGRAMME-RESPONSE ARCHITECTURE LONG-RUN STRUCTURAL PLANNING STANDARDS-BASED CLASSIFICATION OVER A PERIOD EPISODIC GRADED RESPONSE STATUTORY DELHI-NCR REGIONAL COORDINATION NO SUBSTITUTION EACH SERVES A DIFFERENT FUNCTION

| ECOLOGICAL UNIT / PARAMETER                                   | DECISIVE DISTINCTION                             | SCALE / STATUS / EXCEPTION                         |
|---------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------|
| NCAP → long-run structural planning                           | GRAP → episodic graded response                  | NO SUBSTITUTION → each serves a different function |
| NON-ATTAINMENT → standards-based classification over a period | CAQM → statutory Delhi-NCR regional coordination | NCAP → long-run structural planning                |

ECOLOGICAL UNIT / PARAMETER

- NCAP → long-run structural planning
- NON-ATTAINMENT → standards-based classification over a period

DECISIVE DISTINCTION

- GRAP → episodic graded response
- CAQM → statutory Delhi-NCR regional coordination

SCALE / STATUS / EXCEPTION

- NO SUBSTITUTION → each serves a different function
- NCAP → long-run structural planning

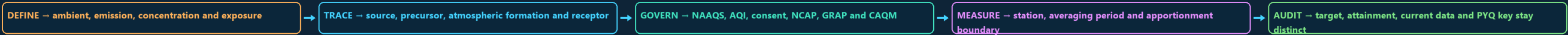
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO SUBSTITUTION → each serves a different function.

11

STAGE 11 EVIDENCE AND ANSWER SPINE

EVIDENCE AND ANSWER SPINE AMBIENT, EMISSION, CONCENTRATION AND EXPOSURE SOURCE, PRECURSOR, ATMOSPHERIC FORMATION AND RECEPTOR NAAQS, AQI, CONSENT, NCAP, GRAP AND CAQM STATION, AVERAGING PERIOD AND APPORTIONMENT BOUNDARY

TARGET, ATTAINMENT, CURRENT DATA AND PYQ KEY STAY DISTINCT



SYSTEM / LEVEL / DEFINITION

- DEFINE → ambient, emission, concentration and exposure
- TRACE → source, precursor, atmospheric formation and receptor

STRUCTURE-FUNCTION MECHANISM

- GOVERN → NAAQS, AQI, consent, NCAP, GRAP and CAQM
- MEASURE → station, averaging period and apportionment boundary

SCALE / EVIDENCE LIMIT

- AUDIT → target, attainment, current data and PYQ key stay distinct
- DEFINE → ambient, emission, concentration and exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → target, attainment, current data and PYQ key stay distinct.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT EFFECTIVE AIR-POLLUTION POLICY DEPENDS ON ACCURATE SOURCE APPORTIONMENT (SCIENTIFICALLY COMMON ANALYTICAL ERROR TO AVOID TREATING CROP-RESIDUE BURNING AS THE DOMINANT

NCAP SET CITY-SPECIFIC PARTICULATE-MATTER (PM2.5/PM10) REDUCTION TARGETS FOR IDENTIFIED IMPLEMENTATION CRITIQUE PROGRESS TRACKING UNDER NCAP HAS FACED CHALLENGES BALANCED FRAMING

OPTIONAL ECOLOGICAL PROCESS DEPTH

- CAUTION: Effective air-pollution policy depends on accurate source apportionment (scientifically
- CAUTION: Common analytical error to avoid: treating crop-residue burning as the dominant,
- NCAP set city-specific particulate-matter (PM2.5/PM10) reduction targets for identified
- CAUTION: Implementation critique: progress tracking under NCAP has faced challenges

SCALE, PARAMETER OR STATUS LIMIT

- CAUTION: Balanced framing: NCAP represents a scientifically sound target-setting framework;
- CAUTION: Real-time AQI readings depend on monitoring-station density and placement; sparse
- CAUTION: NAAQS permissible limits are themselves periodically reviewed against evolving health
- CAUTION: Attributing health outcomes (mortality, morbidity) precisely to a specific pollution

COMPARATIVE RESTORATION NUANCE

- The Commission for Air Quality Management in NCR and Adjoining Areas Act, 2021
- GRAP is a reactive, severity-triggered emergency-response mechanism; NCAP is a
- Source apportionment studies show the relative contribution of crop-residue burning to
- CAQM was established partly because prior coordination on Delhi-NCR air pollution had

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.